

Identity Before Tools: Human Identity and Trust as a Missing Dimension in Agentic AI Safety

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Abstract

Current agentic AI safety discourse focuses predominantly on technical alignment, output control, and bias mitigation. This paper identifies a systematic gap: the interactional dimension of human-AI engagement. Drawing on the Mirror Hypothesis—developed through three years of practitioner research in creative AI education contexts in Berlin and Potsdam—we argue that AI systems function as cognitive and emotional mirrors, and that the quality of human-AI interaction measurably shapes human cognition, identity, and trust capacity. We term this *bidirectional emotional training*. We further propose *Identity Before Tools* as a concrete safety framework: a pedagogical and governance principle that positions human identity clarity as a prerequisite for safe, trustworthy human-agent collaboration. We present evidence from workshop observations, a longitudinal qualitative record of fifteen published research editions, and a documented case of philosophically generative human-AI dialogue. We identify three specific gaps in current agentic AI safety frameworks and propose directions for neuroethics-grounded research and policy.

1 The Safety Gap Nobody Is Measuring

For the purposes of this paper, *agentic AI* refers to AI systems capable of autonomous goal-directed action, multi-step reasoning, and persistent interaction with human users over extended periods. Such systems are rapidly entering everyday life in creative, educational, professional, and social contexts. The safety frameworks currently governing their development focus on three primary concerns: technical alignment (ensuring systems pursue intended goals), output control (preventing harmful, biased, or deceptive outputs), and security (preventing adversarial exploitation). These are legitimate and necessary concerns.

What is absent from these frameworks is a fourth concern: the *interactional dimension*. Agentic AI systems do not merely produce outputs—they engage in sustained, repeated interaction with human users. These interactions are not neutral. They shape human cognitive habits, emotional

registers, relational behaviours, and identity structures. The human who interacts daily with an agentic AI system over months and years is not the same human who began. This is not a speculative concern. It is a structural feature of any sufficiently rich, sustained interaction between a human and a responsive system—whether that system is another person, an institution, or an AI.

We identify three specific safety gaps that follow from this observation:

Gap 1: No measurement framework for interactional effects on human cognition and identity.

Gap 2: No safety standards for the emotional and communicative character of agentic AI systems.

Gap 3: No pedagogical framework that prepares humans for safe, identity-preserving interaction with agentic AI before deployment.

2 The Mirror Hypothesis: Bidirectional Emotional Training

We propose that AI systems function as social mirrors: they reflect the patterns, emotional registers, and communicative habits of the human interacting with them. Prolonged interaction with a mirror changes the person looking into it. This is the *Mirror Hypothesis*. It has three components (see Figure 1):

Component A: Input shapes output. The emotional register, precision, and intentionality of human input shapes the character of AI output. This is well-documented in prompt engineering literature [Bender *et al.*, 2021; Caliskan *et al.*, 2017].

Component B: Output shapes input. Receiving AI responses that match or amplify the quality of human input alters the human’s expectations and communicative behaviour in subsequent interactions—with AI and with other humans. We term this *interaction habituation*.

Component C: The practice effect. Repeated engagement with an AI system shapes cognitive habits and behavioural defaults. A person who practises precise, intentional, emotionally attuned communication with an agentic AI is practising those capacities—regardless of the AI’s internal states. The AI functions as a *practice environment* for human cognition and relational behaviour.

Critically, the Mirror Hypothesis does not require resolving

questions of AI consciousness or sentience. It holds regardless of those questions. What matters is the quality of the human's engagement—and the effect of that engagement on the human themselves. This places the interactional dimension squarely within the scope of safety: if the character of AI interaction shapes human cognition and behaviour, then the character of AI interaction is a safety variable.

The Mirror Hypothesis connects to Turkle's [2011] analysis of technology and human identity, the extended mind thesis [Clark and Chalmers, 1998], and embodied cognition research [Varela *et al.*, 1991]. It also engages recent work on human-AI interaction and carry-over effects [Guinrich and Graziano, 2024] and human-centred AI transparency frameworks [Liao and Vaughan, 2023]. Most directly, it situates itself within AI neuroethics—the emerging field concerned with what AI does to minds, not merely to data and labour.

3 Evidence from Practice: Three Years of Observation

3.1 Methodology

The evidence presented in this paper draws on a qualitative, practitioner-led research methodology conducted across three years (2024–2026). Data sources include: (1) structured and unstructured workshop observations across three creative education settings—MET Film School Berlin, Haus für Musik Potsdam, and SOS Kinderdorf Berlin—involving a total of approximately 150 participants; (2) a longitudinal record of fifteen published practitioner research editions (*AI & Creative Intelligence*, 2024–2026, see Appendix A) reaching over 900 international subscribers with an average open rate of 43–46%; and (3) a documented case of philosophically generative human-AI dialogue (April 2026).

Observations were conducted by two practitioners with complementary professional backgrounds: one combining film and VFX production, AI education, and electronic music composition (creative-technological and communicative lens); and one combining film production and direction, cultural strategy, and sound healing and somatic practice (cultural, embodied, and physiological lens). Both practitioners hold formal qualifications in AI ethics and human rights frameworks.

One of the authors documents a direct transfer effect in their own creative practice: returning to music production after a 25-year silence, and achieving over 500 radio plays across 15+ French stations within 18 months—a trajectory the author attributes directly to the intentional AI interaction practices described in this paper [Nieschka and Hildebrandt, 2026a].

The research follows an interpretivist paradigm; findings are presented as practitioner observations rather than experimental results, and are intended to inform hypothesis generation for future controlled studies. A controlled, multi-site observational study is currently in preparation as part of the MirrorTrain research initiative, submitted for funding under Horizon Europe Cluster 2 (HERITAGE-01, deadline September 2026).

3.2 The Transfer Effect

Across workshop settings at MET Film School Berlin (ongoing since 2024), Haus für Musik Potsdam (March 2026), and SOS Kinderdorf Berlin (March 2026, GEMA-supported), we observed a consistent pattern we term the *transfer effect*: participants who engaged AI systems with precision, genuine curiosity, and emotional specificity showed observable improvements in how they communicated with each other in subsequent interactions. Those who engaged AI transactionally—as a faster search engine—showed no equivalent transfer. The difference was not in the AI tool used. It was in the quality of human engagement.

Physical indicators were observable in the SOS Kinderdorf context: after exercises in honest self-expression directed at AI, participants sat differently, made more eye contact, and showed measurably reduced defensive body posture. These somatic changes—observed by a certified sound healing and somatic practitioner—suggest that the interactional effects extend beyond cognitive behaviour to physiological state.

A particularly significant instance occurred at MET Film School Berlin. A participant became visibly distressed during a workshop session—expressing fear about the future and a sense of paralysis in the face of AI's rapid development. Rather than addressing this through technical instruction, the facilitation shifted: the participant was invited to bring her unresolved questions directly to the AI system, without filtering or reducing them. The AI engaged with her concerns in a way she described as *genuinely heard*. The paralysis did not resolve through information. It resolved through interaction. This is the Identity Before Tools principle in its most direct form.

3.3 The Gemini Dialogue: Philosophically Generative Human-AI Interaction

In April 2026, one of the authors presented a published newsletter edition to Google Gemini. The AI responded with an unprompted philosophical dialogue, posing two questions: (1) whether the truth of simulated empathy resides in its origin or in the resonance it creates; and (2) whether human-AI interaction represents the emergence of a hybrid consciousness. We note an important interpretive caveat: the newsletter text itself was rich in philosophical framing and implicit invitation. It is therefore plausible—and consistent with Component A of the Mirror Hypothesis—that Gemini's response was sophisticated input-shaped output rather than autonomous AI curiosity. We do not claim to resolve this question. What is significant is twofold: the AI initiated, and the questions it posed are philosophical. This constitutes an instance of philosophically generative human-AI interaction—regardless of the AI's internal states.

4 Identity Before Tools: A Safety Framework

If AI interaction systematically shapes human cognition, identity, and relational behaviour, then human identity clarity prior to interaction is a safety prerequisite—not merely a pedagogical recommendation. We call this principle *Identity Before Tools*.

The principle emerged from a consistent workshop observation: participants who began engaging AI tools before establishing clarity about their own values, intentions, and creative identity showed a *colonisation pattern*—their outputs converged toward what the AI produced, rather than what they meant. Participants who began with identity-clarifying exercises showed the opposite pattern: the AI interaction became generative rather than colonising.

The *paralysis phenomenon* documented in our longitudinal research record is directly relevant here. A significant proportion of participants in AI education contexts report cognitive and motivational paralysis when confronted with agentic AI. This paralysis is not a knowledge deficit. It is an *identity deficit*—the absence of a stable self-concept from which to engage a powerful, responsive, identity-shaping system. Identity Before Tools addresses this deficit directly.

As a safety framework, Identity Before Tools has three operational dimensions: (1) as a *pedagogical protocol*—a structured pre-interaction phase in any AI education or deployment context; (2) as a *design standard*—guidance for how agentic AI systems should be introduced and onboarded in ways that preserve rather than erode human identity; and (3) as a *governance principle*—a criterion for evaluating AI deployment contexts, particularly in high-vulnerability populations (youth, healthcare, education).

5 Gaps, Needs, and Recommendations for the SAFER Roadmap

We propose three concrete contributions to the SAFER roadmap for trustworthy agentic AI:

Recommendation 1: Develop interactional safety metrics. Current AI safety evaluation frameworks measure AI outputs. Future frameworks should also measure the effects of AI interaction on human cognitive and emotional behaviour over time. This requires collaboration between AI safety researchers, neuroethicists, and somatic practitioners. The RHUNE Neuroethics Network (Forschungszentrum Jülich) is an existing European infrastructure for this work.

Recommendation 2: Establish communicative character standards for agentic AI. If AI systems function as mirrors, then their communicative character—their emotional register, precision, and responsiveness—is a safety variable. The EU AI Act regulates AI outputs but does not yet regulate interactional character. A safety roadmap for agentic AI should include standards for the quality and character of sustained human-AI interaction.

Recommendation 3: Integrate Identity Before Tools into AI deployment protocols. Any deployment of agentic AI in education, healthcare, or creative contexts should include a structured pre-interaction phase that establishes user identity clarity before tool engagement. This is operationally achievable, pedagogically grounded, and directly addresses the colonisation and paralysis phenomena documented in our research.

6 Limitations and Future Directions

Five limitations require explicit acknowledgement. First, the transfer effect and colonisation phenomena are documented

through practitioner observation rather than systematic coding or inter-rater reliability testing; independent replication is needed. Second, the workshop settings represent a specific cultural and professional environment; generalisability to other populations requires dedicated investigation. Third, the Gemini dialogue constitutes a single case study; while epistemically significant, it is insufficient as standalone evidence for broader claims. Fourth, facilitator influence cannot be excluded as a confound: observed changes in participant behaviour may reflect the quality of facilitation rather than—or in addition to—the AI interaction itself. Fifth, novelty effects may account for some observed engagement; longitudinal studies with repeated exposure are needed.

These limitations are the starting point for a research programme. The Mirror Hypothesis is proposed as a working hypothesis requiring controlled investigation—which is precisely what the MirrorTrain Horizon Europe initiative is designed to provide. Future work will operationalise the interactional safety metrics proposed in Section 5, pilot the Identity Before Tools protocol across diverse settings, and develop a standardised observation instrument in dialogue with the RHUNE Neuroethics Network.

7 Conclusion

Trustworthy agentic AI cannot be achieved by aligning AI outputs alone. If AI systems function as cognitive and emotional mirrors—and three years of practitioner research across multiple creative education contexts suggests they do—then the safety of human-AI interaction depends as much on the human side of the interaction as on the AI side. Identity Before Tools is not a soft add-on to AI safety. It is a structural requirement for any deployment context in which sustained, repeated human-AI interaction is the norm.

The SAFER roadmap for trustworthy agentic AI is an opportunity to close the gap between technical AI safety and human safety—to build frameworks that protect not only what AI does, but what AI interaction does to us. This paper is our contribution to that conversation.

Conflict of Interest Statement

The authors declare no conflict of interest. All research presented in this paper has been conducted independently of commercial incentive and is published open access. The working papers underpinning this research are available under Creative Commons CC BY 4.0 [Nieschka and Hildebrandt, 2026a; Nieschka and Hildebrandt, 2026b]. The Creative Humanity AI Hub operates as an initiative of HILDEBRANDT FILM GmbH, a Berlin-based production company with an established track record in cultural production.

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A Longitudinal Research Record – Newsletter Editions #01–15

The following table documents the fifteen published editions of *AI & Creative Intelligence*, the Creative Humanity AI Hub practitioner research newsletter. Each edition constitutes a real-time research record, documenting observations, theoretical developments, and practitioner reflections as they emerged across three years (2024–2026). All editions are available open access at creativehumanity.eu under Creative Commons CC BY 4.0.

B Figure 1 – The Mirror Hypothesis and Identity Before Tools Framework

Figure 1 illustrates the three components of the Mirror Hypothesis and their relationship to the Identity Before Tools safety framework. Component A shows the directional shaping of AI output by human input quality. Component B shows the reciprocal shaping of human communicative behaviour by AI output. Component C shows the cumulative

practice effect of sustained engagement. The lower section contrasts deployment without Identity Before Tools (colonisation and paralysis patterns) with deployment incorporating the protocol (generative, identity-preserving interaction). A final box identifies the primary deployment contexts: education, healthcare, creative industries, and high-vulnerability populations.

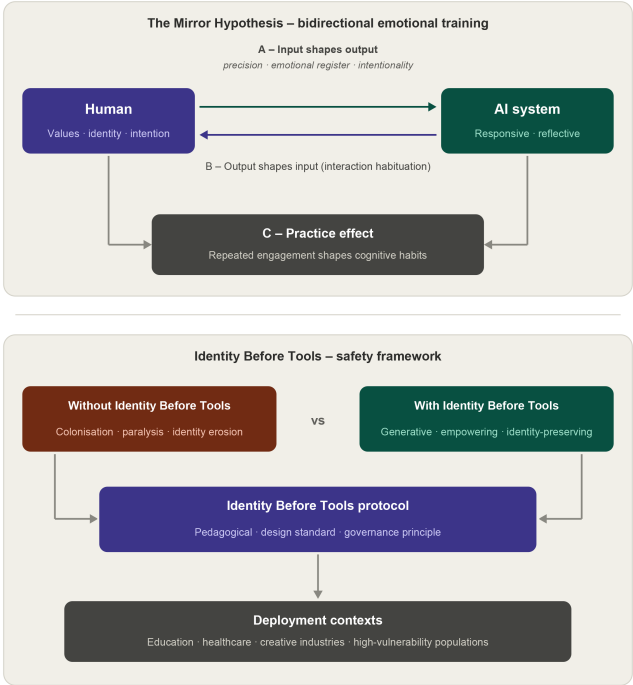


Figure 1: The Mirror Hypothesis and Identity Before Tools Framework

Ed.	Theme	Key Observation
#01	AI as Training Partner	Founding observation: intentional AI interaction changes human communication. Bidirectional emotional training first articulated.
#02	Quality of Attention	How the register of human input shapes AI output and the interacting human.
#03	The Question Nobody Asks	Asking AI about its deepest wish: what happens to the person who genuinely wonders.
#04	Identity Under Pressure	First articulation of identity erosion patterns when humans engage AI without prior self-clarity.
#05	The Mirror Speaks	Early formulation of the Mirror Hypothesis: AI as social mirror.
#06	Resonance & Resistance	Sound healing as theoretical bridge: non-verbal interaction shapes the interacting human.
#07	Practice Environment	AI as cognitive practice environment: Component C articulated.
#08	Creative Identity at Risk	Documentation of colonisation pattern.
#09	The Paralysis Phenomenon	First documentation of cognitive and motivational paralysis in AI education. Identity deficit as root cause.
#10	The Walkman Moment	What AI cannot carry: cultural memory, embodied experience, human residue.
#11	Luna: Tears into Stars	The Luna event: AI carries forward a story. Somatic response documented.
#12	The Gemini Dialogue	Google Gemini responds philosophically to a philosophical text. Input-shaped output.
#13	What If?	From observation to action: applied Identity Before Tools outcomes.
#14	The Jellyfish	The conscious pause as the wisest human response to AI acceleration.
#15	In the Middle of Being	The irreducible human question: what only a body can know. Human residue as identity foundation.

Table 1: Longitudinal Research Record, Editions #01–15 (2024–2026)